Leveraging Textual Description and Structured Data for Estimating Crash Risks of Traffic Violation: A Multimodal Learning Approach

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***Abstract*— This study introduces a novel methodology that integrates both structured data and unstructured violation descriptions, addressing a critical gap in current crash risk estimation techniques. By combining these two data types, our approach captures a more comprehensive picture of violation characteristics, enabling more precise identification of crash-prone violations. We propose an innovative framework that leverages advanced Natural Language Processing (NLP) techniques alongside state-of-the-art Large Language Models (LLMs) to convert raw textual information into actionable features, thereby equipping law enforcement with a powerful tool to systematically assess crash risk and prioritize high-risk violations for targeted interventions. Specifically, we compare four NLP algorithms—Jaccard, TF-IDF, fastText, and BERT—with an LLM framework (GPT-3.5) to process violation descriptions. Following text conversion, we implement eight data-driven classification models, ranging from linear and tree-based to neural network approaches, to predict crash-prone violations. Our experimental results reveal that BERT and GPT-3.5 significantly improve classification accuracy and recall by extracting meaningful insights from text, whereas the other NLP methods may introduce noise. Among the classification models, TabNet outperforms others with an accuracy of 0.9717, recall of 0.8861, and AUC of 0.9658.**

***Index Terms*— Crash risk evaluation, traffic violation, natural language processing (NLP), attention mechanism, large language models (LLM).**

I. Introduction

TRAFFIC crashes continue to impose significant human and economic costs, with traffic violations recognized as a primary contributing factor according to the National Highway Traffic Safety Administration (NHTSA, [1]). As a result, accurately estimating the crash risk associated with different violations is paramount for effective enforcement and timely risk reduction, especially in mitigating severe injuries and fatal outcomes [2], [3]. In enforcement datasets [4], Traffic violations are documented both as textual descriptions and as structured variables, including time, location, driver demographics, and vehicle details. This practice poses challenges for systematic analysis because traditional risk estimation models have predominantly relied on structured information [5], [6], [7], [8]. Although these structured variables offer valuable insights into the circumstances of traffic incidents, they may not fully capture the contextual information inherent in narrative descriptions.

Textual descriptions often contain critical details about the manner in which a violation occurred. These elements can be pivotal in understanding why certain violations are more likely to lead to crashes [9]. For instance, while two violations might share similar structured characteristics, the specific context provided in the narrative may reveal differences in external factors that affect the incident's severity. Incorporating insights from textual descriptions deepens our understanding of the contextual interplay underlying traffic violations. Supplementing structured data with narrative details can uncover key elements that elevate crash risk, revealing previously overlooked risk contributors. This enriched perspective supports the development of more informed and proactive enforcement strategies.

To explore this potential, this study develops a data-driven framework that integrates unstructured textual descriptions with structured datasets to estimate crash risk at the individual violation level, defined as the probability that a violation leads to a crash. We employ Natural Language Processing (NLP) and Large Language Models (LLMs) to extract structured representations from textual violation descriptions, allowing them to be effectively combined with traditional structured variables. To improve violation risk estimation, we adopt TabNet, an attention-based deep learning model that enables dynamic feature selection and enhances interpretability. By selectively focusing on the most relevant features at each decision step, TabNet effectively captures complex interactions between structured data and extracted textual features. This approach allows for a more adaptive and context-aware assessment of how different violations pattern contribute to crash risk while maintaining computational efficiency. By leveraging multi-modal learning, which integrates both structured and unstructured data, this study explores an effective approach to integrating textual data into predictive models and refining risk assessment. The proposed framework also offers a suite of practical tools to enhance road safety and refine law enforcement strategies. By leveraging deep learning techniques, it provides actionable insights that allow agencies to precisely pinpoint high-risk violations. This, in turn, enables more effective resource allocation

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and targeted interventions to address specific traffic safety challenges.

II. Literature Review

In the past decades, research about traffic violations mainly focused on spatial-temporal correlation and factor analysis [3]. [10] collected traffic fatality records among more than 300 cities in China. The conclusion demonstrated that high-risk driving behavior in urban areas reflected a strong spatial relationship and core-periphery structure. Based on the structured speeding violation data, the speeding violation density map of heavy vehicles was created by spatial analysis [11]. Moreover, with the help of automated enforcement system data and a spatiotemporal kernel density estimation model, the spatiotemporal patterns of traffic violation hotspots and coldspots were identified [12]. As to the temporal correlation, traffic violations were more likely to happen at midday and during weekends [5]. The multinomial logit model [13] was applied in the factor analysis, and the results revealed that congested and severe weather conditions were the leading risk factors causing traffic violations. Reference [2] indicated that novice drivers, males, higher quality vehicles, overload, and poor visibility are more likely to be related to traffic violations. Additionally, vehicle types, vehicle ownership, and driving experiences have potential impacts on the severity of traffic violations [14]. Despite these advancements, a key limitation is that while existing studies identify violation patterns and contributing factors, they do not establish a direct connection between specific violations and crash risk. Reference [3] attempted to address this by employing spatial analysis to examine the relationship between violations and crash frequency at an aggregate level. However, it does not capture the relationship at the individual case level. Without such granularity, it is challenging to determine how specific violations contribute to crash occurrences, limiting the precision of enforcement strategies. Bridging this gap is essential for designing targeted interventions, improving risk assessment, and ultimately enhancing road safety outcomes.

Existing crash models can be divided into three main categories: traditional statistical models, advanced statistical methods that further reduces unobserved heterogeneity, and data-driven approaches (aka machine learning/data mining) [15], [16]. Each category presents trade-offs between predictive accuracy and the ability to understand underlying causality or maintain interpretability. Traditional statistical models commonly include binary/multinomial logistic regression, negative binomial regression, and Poisson-lognormal among others [16]. However, these models rely on fixed parameters and are often limited by issues such as endogeneity and unobserved heterogeneity [17], which can undermine both their predictive accuracy and the reliability of the insights they provide [18]. To address these limitations, advanced statistical models have been developed, such as random parameters, latent-class, and multi-parameter (e.g., negative binomial-Lindley) models, which overcome some of the restrictions of traditional statistical approaches [17], [18]. The extensive testing and modification of various hypotheses during the modeling process can introduce subjectivity and bias from safety analysts. Therefore, existing studies [19], [20], [21], [22] have proposed an optimization-driven approach that fully accounts for grouped random parameters, heterogeneity in means, variable transformations, multicollinearity, and diverse random parameter distributions to accurately identify the critical safety factors. However, the increased complexity of these models renders their estimation both challenging and time consuming and more sophisticated models are not necessarily better models [16].

Data-driven approaches have been boosted by advancements in Artificial Intelligence. These methods excel at handling large datasets and capturing complex, context-dependent interactions among observed variables, thereby achieving improved predictive performance [15], [23], [24], [25]. Moreover, data-driven approaches are not limited to structured data, such as the tabular datasets commonly used in traffic safety. They can also process unstructured data from multiple modalities, such as textual descriptions or video images, to enrich analyses and provide more comprehensive insights. For example, existing studies have analyzed text-based data using topic models [26] or the Apriori algorithm for association rule mining [27], several recent studies [9], [28], [29] have leveraged advanced machine learning and deep learning algorithms to process unstructured datasets and deliver practical tools for enhancing traffic safety.

While significant advancements have been made in understanding the patterns of traffic violations and crash models, the interconnected nature of traffic violations and crashes is often overlooked. For example, studies may analyze violation patterns and crash occurrences separately without fully exploring how specific violation behaviors contribute to subsequent crash risks in a dynamic traffic environment. Considering recent advancements in AI, such as Natural Language Processing (NLP) and Large Language Models (LLMs), we aim to fully leverage the information in violation records—both structured data and textual descriptions—to enhance the estimation of individual violation risk (i.e., the probability that a violation will lead to a crash) and support law enforcement decision-making and support law enforcement decision-making. Data-driven approaches excel in this task by identifying complex patterns within the data, enabling more precise risk assessments [28], [30], [31], [32]. Most data-driven crash prediction models for binary outcomes rely on tree-based methods, but these often require extensive feature engineering to achieve the desired predictive performance. However, emerging deep learning architectures—particularly those that utilize attention mechanisms—have demonstrated superior performance on various tasks compared to traditional models [33]. The attention mechanism effectively highlights important features and interactions, thereby reducing the need for manual feature engineering and enhancing overall model interpretability.

Therefore, the contributions of this study are twofold. First, we comprehensively compared various NLP algorithms and LLM frameworks to identify the most suitable approach for embedding textual descriptions into structured datasets, thereby enhancing the predictive performance in identifying violations that lead to crashes. Second, we deploy a deep

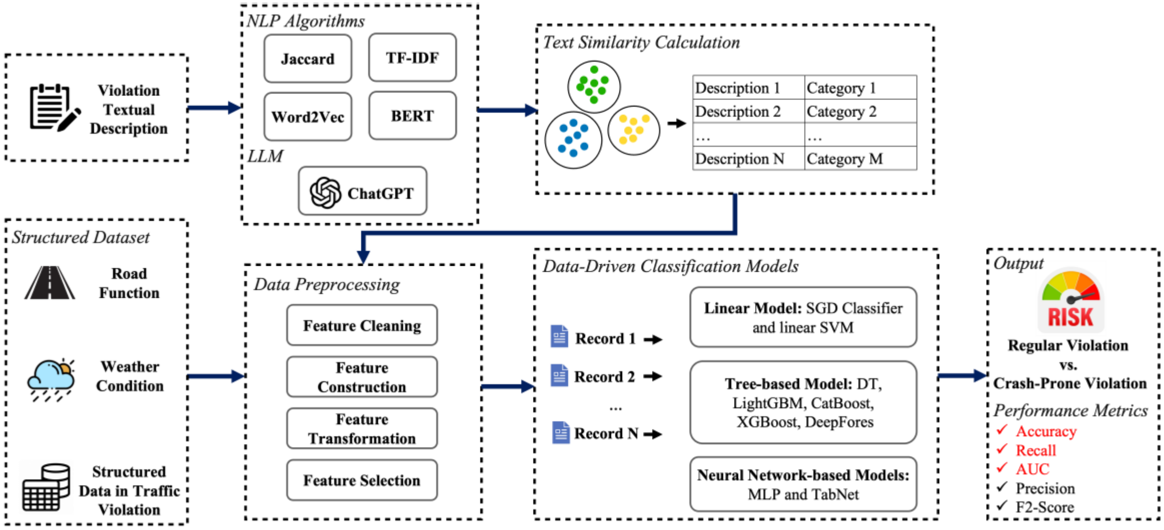

Fig. 1. Framework of the crash-related traffic violation prediction.

learning framework with attention mechanisms (e.g., TabNet, [34] to further improve predictive performance and enhance model interpretability. Note that the interpretability discussed here differs from the causal understanding offered by advanced statistical models, as detailed in Section IV-C. This paper aims to provide law enforcement agencies with a practical tool that shifts assessments from traditional, subjective judgments to data-driven, reliable evaluations.

III. Methodology

A. Problem Formulation

The research problem is formulated into two parts: semantic similarity calculation of violation description and classification of violations leading to crashes. In the traffic violation data, the description records the details of each violation. For example, the driver failed to obey the properly placed traffic control device. In this study, we mainly extract the violation types from the text description because i) it is the primary information in the description, and 2) it is most frequently associated with crashes can inform targeted interventions and policy decisions to improve road safety [5]. Refer to [35], there are ten types of traffic violations in total, including the issue of documentation, disobeying traffic control devices, unsafe speed, issue of vehicles, reckless driving, driving under the influence (DUI), distracted driving, failure to yield, illegal parking, and failure to wear a safety belt. In order to accurately match the category of each traffic violation based on the description, part of descriptions is manually labeled as reference. Then, the rest of the unannotated descriptions are automatically notated by NLP algorithms or LLM based on semantic similarity. In the second step, the processed description information and structured data are concatenated to train the binary classification model (i.e., predicting whether a violation will lead to a crash) to distinguish crash-related violations from all traffic violations. **Figure 1** shows the flow chart of the proposed framework. The detailed procedures of data processing, NLP algorithms, LLM and the TabNet are discussed in the following sections.

B. Dataset Overview

This study uses traffic violation data from 2020 to 2022, along with information on weather and road functional class in Montgomery County, Maryland, United States. In the dataset, the crash involvement of traffic violations is recorded, enabling the analysis of the crash risk associated with traffic violations.

1) Traffic Violation Data: The traffic violation dataset is collected by the police department in Montgomery County. All electronic traffic violations issued in the county are recorded, including the date, time and location, description, safety-related information, drivers, and vehicle information of each violation. The raw dataset has 1,792,091 records from 2012 to 2022. The priority of this paper is the violations caused by risky driving behavior and to provide reference to the police on patrol. However, the violations with a search conducted are highly related to contraband and criminal activity, which is not the focus of this paper. Some objective arrest types, such as marked Laser and license plate recognition, are not involved in further analysis. In other words, only violations that did not involve a search conduct and were influenced by subjective law enforcement (e.g., marked and unmarked patrol) are retained. Then, the records containing missing values are removed, and only the records that occurred after 2020 are selected because the more recent the data, the more complete it is. Finally, there are 69,209 records used for this study, as shown in **Figure 2.** Note that there are only 2,055 distinct violation descriptions because many violations share the same description.

Moreover, there are multiple steps in this study adopted for feature cleaning, construction, and selection purposes and these steps are described as follows:

- Filled the missing information of each weather variable with its average.
- Removed unrelated variables such as year, make, model of the vehicles, etc.
- Created a new variable, "In State", containing more comprehensive information. There are two attributes, Yes and No, in the variable, by fusing the location of violation

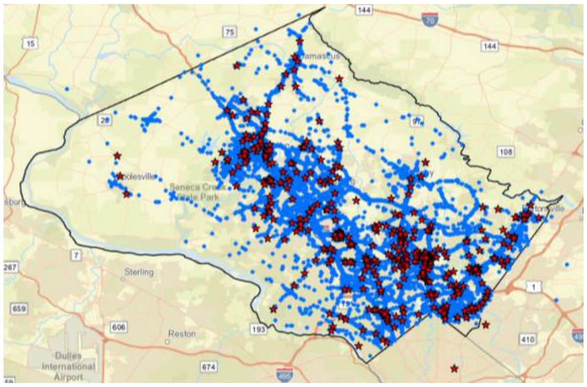

Fig. 2. Map of all violations in montgomery county from 2020 to 2022 (Black line, blue dot, and red star represent county boundary, regular violation, and crash-related violation, respectively).

and state of driver's license. Yes in In State is defined as the state where the violation occurred is the same as the state where the driver's license was issued.

- Combined redundant attributes into certain attributes. For example, there are different types of vehicles, including but not limited to automobiles, station wagons, light-duty trucks, heavy-duty trucks, transit buses, school buses, and so on. These attributes were combined into five categories: passenger car, truck, bus, motorcycle, and others. The same process is also applied to the color of vehicles.
- Label-encoded the categorical variables and scaled the continuous variables to a similar range.
- Employ variation inflation factors to assess multi-collinearity, set a threshold of 5, and eliminate any variables exceeding the threshold [36].
- Processed the violation description into the type of violation by NLP algorithms. The details are introduced in Section III-C.

2) Weather Condition and Road Functional Class Data: Weather condition data are collected from the Automated Surface Observing System (ASOS) of the National Centers for Environmental Information (NOAA). Each traffic violation event is assigned to an ASOS weather station closest to it in Maryland. Weather condition data, including precipitation (inches), visibility (miles), temperature (Fahrenheit), and wind speed (knots), is collected for the specific hour when the traffic violation event happened.

Roadway functional class data are collected from the Maryland highway performance monitoring system roadway functional classification map. It contains GIS shapefile linear features showing the functional classification of the public roadways in the state of Maryland. Since, in some of the traffic violation events, the drivers were pulled over on major roads while they stopped their vehicles on minor roads. The GPS location reported by the police officers is where the vehicles stopped. To match these violation events to the major roads where they happened, this study only considers roadways with the functional class from one to six, representing interstate, freeway, principal arterial, minor arterial, major collector, and minor collector, respectively. Each traffic violation is then matched with the roadway segment that is closest to it. Finally, the categorical and numerical features of this study are shown in **Tables** V and VI in Appendix.

C. Natural Language Processing (NLP) Algorithms

NLP algorithms are employed to convert descriptions of traffic violations into categorical types. To ensure the accurate categorization of violation types by the NLP algorithms, we initially manually annotate the top J descriptions with corresponding traffic violation categories, where $J = 200$ in this study. For example, the types "failure to obey properly placed traffic control device instructions" and "using hands to operate a handheld telephone while the vehicle is in motion" are categorized as "disobeying traffic control device" and "distracted driving," respectively.

Each unannotated description u_i is then compared with all annotated descriptions $A = \{a_1, a_2, \ldots, a_j\}$ to calculate similarity. Two similarity metrics are adopted in this study: Jaccard and cosine similarity. Take Jaccard similarity as example, it is calculated based on common words in two text corpora. After iteration, each unannotated description u_i can be represented as a vector $s_i \in \mathbb{R}^J$, where each value in the vector represents the similarity between a type of traffic violation from the annotated descriptions and the unannotated description. By averaging the similarities from the K categories ($K = 10$), we obtain a categorical vector $p_i \in \mathbb{R}^K$. Finally, the category with the highest average similarity p_i^* is assigned to the unannotated description u_i. Through the procedure in **Table I**, we can convert textual descriptions of violations into violation types. The calculation of cosine similarity is similar, but it begins with sentence embedding, which applies NLP algorithms to convert textual descriptions into numerical representations. The subsequent steps to structure violation text descriptions into types are the same as for Jaccard similarity but are based on the output of sentence vectorization. It is important to note that different NLP algorithms generate distinct vectorized representations for the same textual description. However, LLMs differ from traditional NLP methods as they do not rely on predefined embeddings and can directly output the most suitable classification. This detail is introduced in Section III-C.4. The appropriate approach is determined by comparing the performance of various NLP algorithms in Section IV.

1) Jaccard Similarity: Jaccard similarity is defined as the size of the intersection divided by the size of the union of the two sets, as shown in Equation 1. It measures text similarity by calculating the common words used in two corpora. It is the most fundamental and efficient way to calculate text similarity. However, the method's performance is inclined to decrease when the corpus covers various contents and the size of the text rises.

$$J(A, B) = \frac{\left|A \bigcap B\right|}{\left|A \bigcup B\right|} \tag{1}$$

where A and B are two corpora, $\left|A \bigcap B\right|$ and $\left|A \bigcup B\right|$ represent the size of their intersection and union, respectively.

2) Bag of Words (BOW) and Term Frequency-Inverse Document Frequency (TF-IDF): How to convert sentences into

TABLE I
PROCEDURE OF CONVERTING TEXT DESCRIPTION TO VIOLATION CATEGORY

Input: Annotated description set $A = \{a_1, a_2, \dots, a_j\}$, Unannotated description set $U = \{u_1, u_2, \dots, u_i\}$, Violation categeory $C = \{c_1, c_2, \dots, c_k\}$, NLP algorithm $vec(\cdot)$, and similarity measure $sim(\cdot)$

For u_i in U:
    For a_j in A:
        $s_{i,j} = sim(vec(u_i), vec(a_j))$
        $s_i = [s_{i,1}, s_{i,2}, \dots, s_{i,j}] \in \mathbb{R}^j$
    $p_{i,k} = \frac{1}{|c_k|} \sum_{a_j \in c_k} s_{i,j}$
    $p_i = [p_{i,1}, p_{i,2}, \dots, p_{i,k}] \in \mathbb{R}^k$
    $p_i^* = \arg\max_k (c_{i,k})$

Output: assigned the violation category (p_i^*) for each unannotated description

vectors to make the model understand the textual information is an essential issue in NLP. There are several ways to do so: BOW (bag of words), TF-IDF (term frequency-inverse document frequency), and Word Embeddings (Word2Vec, Glove, and fastText).

BOW [37] generates a collection of vectors with the count of word occurrences or term frequency (TF) in the text. However, it does not consider the order between words and reflect the importance of word solely on the term frequency. Thus, pronouns and stop words tend to have a higher weight, but these high-frequency words are not the key information in the document. As an improvement, the TF-IDF introduces inverse document frequency (IDF) to weaken the weight of pronouns and stop words. The higher the document frequency, the lower the weight of this term. The similarity is measured by the TF-IDF vectors, which consist of TF-IDF weights of each word from all descriptions [38].

3) Word Embeddings: Word Embeddings create one vector per word by pre-trained NLP models. For example, Word2Vec uses shallow 2-layer neural networks to ingest a corpus and produce sets of vectors. It consists of two major steps, the continuous bag of words (CBOW) and the Skip-gram. The CBOW method's core idea is to key a word out of a sentence and use the preceding and following words to predict the keyed-out word. Skip-gram contrarily uses a word as input and asks the network to predict its contextual words. FastText [39] uses a similar CBOW structure, but fastText assumes n-grams of characters form a word, where n could range from 1 to the length of the word. FastText uses character-level n-grams of each word as input, averaging them to get the document's representative vector. N-grams help to find the vector representation for rare words and words not present in the dictionary (Out-of-vocabulary words) [49]. The word vectors' distance can be used to measure the semantic similarity between words. The 2M vocabulary from CommonCrawl [42] is used as word embeddings to pre-train the fastText model and to obtain the word vector to calculate the semantic similarity in this study.

4) Bidirectional Encoder Representations from Transformers (BERT): BERT is a stack of multiple layers of transformer encoder [33]. BERT is a bidirectional model based on the transformer architecture, which replaces the sequential nature of RNN (LSTM & GRU) with a much faster attention-based approach. Attention is a deep learning model that mimics the human attention mechanism, differentially weighting the significance of each part of the input data. Self-Attention means attention comes from the source sentence and the word vectors themselves. The transformer is an overlay of the self-attention model. The attention mechanism can better solve the long-range sequence dependence problem and has parallel computing capability.

BERT is pre-trained on two unsupervised tasks, Masked Language Model and Next Sentence Prediction [43]. These allow us to use a pre-trained BERT model by fine-tuning downstream specific tasks such as sentiment classification, intent detection, question answering, and more. BERT representations are dynamic contextual embeddings in which the vector for each word is different in different contexts. On the other hand, Word2vec or fastText embeddings are static embeddings, meaning that the method learns one fixed embedding for each word in the vocabulary [44]. Note that using the vectorized representation generated by BERT directly in semantic similarity calculations via cosine similarity may lead to unsatisfactory results [45], [46]. The main reason is that sentence vectors generated by BERT are not on a standard orthogonal basis, but the cosine similarity calculation is in a linear space [46]. A feasible approach to address this issue is to whiten (i.e., normalization) the vectorized representation from BERT [47]. After whitening, the features of the sentence vector are converted into independent and identical distributions, and then the cosine similarity is useable for the whitened vectorized representation of each violation description. Specifically, the vectorized representation of all violation descriptions from BERT should be whitened first, and then calculate the cosine similarity to assign the type of violation to each description.

5) Large Language Models (LLMs): LLMs are advanced AI systems trained on vast amounts of text data to understand and generate human-like text [48]. They use deep neural networks, particularly transformer architecture, to convert text into high-dimensional vectors, enabling them to capture complex language patterns and semantics. The scale and complexity of LLMs have expanded exponentially, now encompassing models with hundreds of millions or even billions of parameters. This growth in model size is directly related to enhanced performance across a range of NLP tasks. ChatGPT [49] is a prominent example of a LLM. The effectiveness of ChatGPT, like other LLMs, heavily relies on the quality of prompts provided by users. These prompts serve as the model's input, guiding its output generation and ensuring that the responses are contextually relevant and aligned with the user's intent. Therefore, crafting well-defined and informative prompts is crucial for leveraging the full potential of ChatGPT in various applications [48]. Unlike the aforementioned NLP models that calculate similarity, we directly utilize ChatGPT with meticulously crafted prompts to guide its understanding of

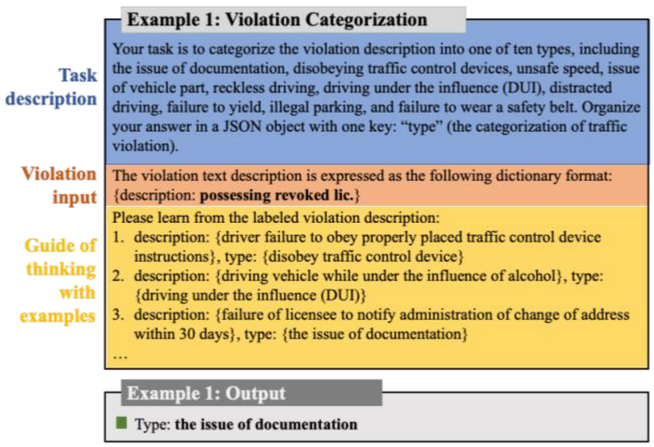

Fig. 3. Example of complete prompts for traffic violation categorization.

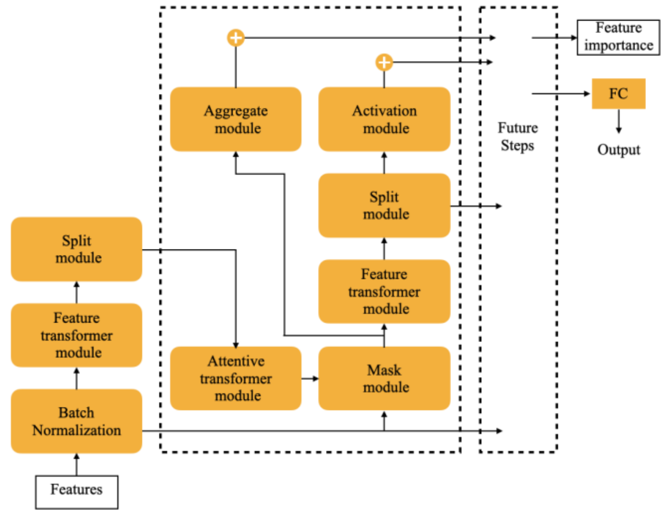

Fig. 4. TabNet encoder structure [34].

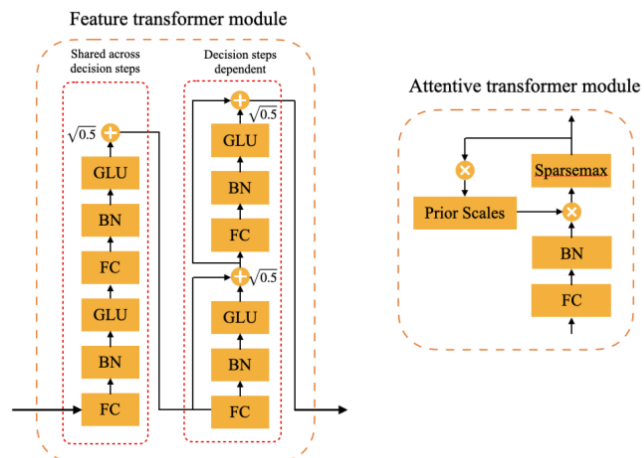

Fig. 5. Feature transformer module and Attentive transformer module structure [34].

textual descriptions and subsequently categorize them into traffic violation types. Typically, ChatGPT is capable of zero-shot learning, meaning it can perform tasks without explicit training or specific examples [50]. It utilizes its extensive pre-training on diverse datasets to generate responses. To enhance ChatGPT's performance and ensure accuracy while avoiding hallucinations, we provide a small number of examples to guide its responses, a technique known as few-shot learning [51]. The detailed Prompt design is shown in **Figure 3**. The specific ChatGPT employed in this study is GPT 3.5 (version gpt-3.5-turbo with 20 billion parameters) from OpenAI.

D. TabNet

In the landscape of tabular data prediction, tree-based models are renowned for their outstanding performance, computational efficiency, and interpretability. However, they often require time-consuming feature engineering to achieve optimal results. On the other hand, deep neural networks (DNNs) excel in handling sequence or unstructured data due to the capacity for representation learning, with the added advantage of requiring less manual feature engineering. TabNet [34] combines the advantages of the two. It incorporates the end-to-end learning and representation capabilities of DNNs, reducing the need for extensive feature engineering, while also integrating the interpretability and sparse feature selection of tree-based models. This unique blend makes TabNet a compelling alternative to traditional tree-based models for tabular data [52]. TabNet implements a sequential multi-step framework to construct a neural network similar to an additive model and achieve feature transformation and selection by encoder and decoder. As shown in **Figure 4**, TabNet transforms input data and features sequentially at the batch level instead of doing it globally. For instance, normalizing numerical features at the batch level is also called batch normalization. The same $B \times N$ dimension batch is fed into the encoder, where B is the batch size, and N is the number of dimensions of the feature. In the encoder, the processed batch is forwarded to the attentive transformer module, split module, mask module, and feature transformer module at each step.

The feature transformer module incorporates two layers: the feature sharing layer (sharing parameters in all rounds of decision making) and the feature independent layer (non-parameter sharing), shown in **Figure 5**. The former computes feature commonality, and the latter computes feature's unique characteristics. In both layers, batch-normalized features $X \in R^{B\times N}$ are connected to a fully connected layer (FC), gated linear unit (GLU) activation, and eventually to a weighted residual normalized connection. The weight $\sqrt{0.5}$ helps stabilize the learning process by ensuring a slight sharp variance fluctuation throughout the network.

Then the output goes to the split module, which splits the vector output from the feature transformer layer into two parts for the decision step, as shown in Eq. (2).

$$[d[i], a[i]] = f_i(M[i] \cdot f) \tag{2}$$

where $d[i] \in R^{B\times N_d}$, $a[i] \in R^{B\times N_a}$, $M[i] \in R^{B\times D}$. $d[i]$ is fed into the activating function and used to calculate the final output of the model. $a[i]$ is used as the next step's input

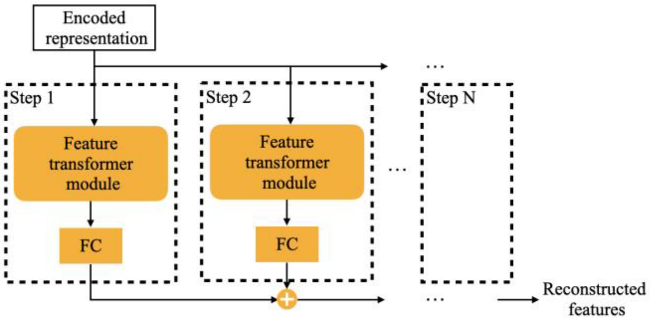

Fig. 6. TabNet decoder structure [34].

to calculate the Mask module for the next step's Attentive transformer module.

The learnable mask $M[i] = argmin_{\{p\in\Delta\}} ||p - P[i-1]\cdot h_i(a[i-1])||$, where $P[i] = \prod_{j=1}^{i}(\gamma - M[j])$, $\Delta^{\{K-1\}} = \{\boldsymbol{p} \in R^K | \mathbf{1}^T\boldsymbol{p} = 1, \boldsymbol{p} \geq \mathbf{0}\}$. $P[i]$ is the prior scales term, which indicate the extent to a feature has been used in previous steps. If a feature has been used many times in previous steps, it should not be selected by the model anymore, so the model reduces the weight of such features by $P[i]$. Based on intuition, $\gamma = 1$ indicates that each feature can only be utilized once at each step. The restriction on feature reusing loosens as γ increases [53]. $M[j]$ can be interpreted as the attention weight assignment of the model for the batch samples at the current step. It is worth noting that the attention weights output by the attentive transformer layer is different for different samples, an attribute called instance-wise [34]. The attribute is not achievable for the tree model.

The decoder consists of a feature transformer module at each step, as shown in **Figure 6**. The decoder consists of a feature transformer layer. The decoder is to reconstruct the representation vector into the feature and then similarly sums over several steps to get the final reconstructed features. Consider a binary mask module $S \in \{0,1\}^{B\times D}$, the decoder produces the reconstructed features when the TabNet encoder receives the input $(1-S)\hat{f}$. In addition, for the model to learn the representation of the entire feature data, not just certain features, the matrix S is resampled from Bernoulli distribution in each round during the training process of self-supervised learning to ensure the overall representational power of the encoder model.

TabNet has great interpretability. Meanwhile, TabNet eliminates instance-wise redundant features and improves global feature selection. Through learning mask, TabNet performance is close to global feature selection. $\eta_b[i]$ denotes the aggregate decision contribution at i^{th} decision step for the b^{th} sample. Intuitively, if the output is negative, then all features at i^{th} should have zero contributions to the overall decision.

$$\eta_b[i] = \sum_{c=1}^{N_d} ReLU(d_{b,c}[i]) \tag{3}$$

Then it is clear from the previous discussion that the global importance of the normalized feature importance can be expressed as

$$\psi_{b,j} = \frac{\sum_{c=1}^{N_d}\eta_b[i]M_{b,j}[i]}{\sum_{j=1}^{D}\sum_{i=1}^{N_{steps}}\eta_b[i]M_{b,j}[i]} \tag{4}$$

In summary, TabNet first employs a self-supervised learning approach to obtain the representation of tabular data through the encoder-decoder framework, then uses the embedded features to do the classification and regression tasks. The process is similar to the pre-training and fine-tuning processes in NLP tasks. Self-supervised learning acts as a data augmentation process. TabNet can achieve high performance even when there are relatively few samples with labels or imbalanced data. Also, the interpretability of the tree model is well combined with the representation power of a deep natural network.

IV. Result and Discussion

This section begins by evaluating the validity of violation descriptions using four NLP algorithms and a LLM framework in Section III. Next, we select eight data-driven approaches as candidates to showcase the performance of the proposed framework. These include linear models such as the Stochastic Gradient Descent (SGD) classifier and Support Vector Machine (SVM) with a linear kernel. For tree-based models, we considered Decision Tree (DT), Light Gradient Boosting Machine (LightGBM), Categorical Boosting Algorithm (CatBoost), Extreme Gradient Boosting (XGBoost), and a deep tree-based ensemble model (DeepForest). Additionally, we included neural network-based models, specifically the Multi-Layer Perceptron (MLP).

The processed dataset (**Tables V** and **VI**) is split into training, validation, and testing sets by 8:1:1. The experiments are implemented in Python using PyTorch and scikit-learn libraries in Google Colab pro. To comprehensively evaluate the predictive performance, accuracy, recall, precision, F2-score, and Area Under Curve (AUC), calculated with **Eqs. (5)-(8)**, are selected. In this study, positive is defined as violations leading to crashes, and negative is a regular violation, defined as one that does not lead to a crash. The TP, FP, TN, and FN represent true positive, false positive, true negative, and false negative, respectively. AUC is the total area under the Receiver Operator Characteristic (ROC) curve. It is used as a critical measurement to present the ability of a binary classification model, especially for imbalanced data. The accuracy represents the percentage of violations correctly identified, whether crash-related or regular. The definition of recall is the percentage of crash-related violations classified correctly among all real crash-related violations. The denominator of precision is the crash-related violations that the model predicts, and the numerator is the same as recall. F_β score can put different emphasis on recall and precision by β. For a highly imbalanced dataset, the precision could be sacrificed to obtain better recall, particularly in traffic safety, because transportation agency expects more crashes to be predicted correctly. The F2 score is an example of the F_β score with a beta value of 2.0. It has the effect of lowering the importance

TABLE II
PERFORMANCE OF DEEPFOREST UNDER DIFFERENT NLP ALGORITHMS

Performance Metric	DeepForest				
	Accuracy	Recall	Precision	F2 score	AUC
Without	0.9896	0.7672	0.8753	0.7867	0.8819
Jaccard	0.9899	0.7601	0.8914	0.7832	0.8786
TF-IDF	0.9896	0.7648	0.8774	0.7850	0.8807
fastText	0.9895	0.7625	0.8747	0.7825	0.8795
GPT-3.5	0.9905	0.7767	0.9051	0.7994	0.8871
BERT	**0.9908**	**0.7814**	**0.9106**	**0.8042**	**0.8895**

TABLE III
PERFORMANCE OF TABNET UNDER DIFFERENT NLP ALGORITHMS

Performance Metric	TabNet				
	Accuracy	Recall	Precision	F2 score	AUC
Without	0.9666	0.8772	0.4988	0.7616	0.9541
Jaccard	0.9686	0.8028	0.4957	0.7143	0.9428
TF-IDF	0.9714	0.7284	0.5596	0.686	0.9047
fastText	**0.9769**	0.8084	**0.5964**	0.7515	0.9526
GPT-3.5	0.9690	0.8692	0.4934	0.7543	0.9595
BERT	0.9717	**0.8861**	0.5459	**0.7878**	**0.9658**

of precision and increasing the importance of recall.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{5}$$

$$Precision = \frac{TP}{TP + FP} \tag{6}$$

$$Recall = \frac{TP}{TP + FN} \tag{7}$$

$$F_\beta = (1 + \beta^2)\frac{Precision \times Recall}{\beta^2 \times Precision \times Recall} \tag{8}$$

A. Ablation Experiment

Two kinds of classification models, including tree-based models and neural networks under four NLP algorithms (i.e., Jaccard, TF-IDF, fastText, and BERT) and a LLM framework (i.e., ChatGPT), are compared to examine the validity and adaptability of information from violation description on prediction. The baseline is the model without textual description involved. DeepForest [54] is an advanced tree-based model, which could achieve better feature representation and learning performance by cascading multiple layers of random forest. Therefore, it is selected as the representative of the tree-based model. In **Table II**, it can be found that DeepForest achieves a decent performance under all kinds of NLP algorithms. Models without description information perform better than Jaccard, TF-IDF, and fastText in terms of recall, F2 score, and AUC. It indicates that inappropriate extraction of violation descriptions adversely affects classification performance. However, when GPT-3.5 and BERT are applied for text processing, the model significantly outperforms its counterpart which lacks textual description across all five metrics. This suggests that proper handling of violation descriptions via NLP algorithms has the potential to enhance the model's performance in identifying crash-related violations. GPT-3.5 and BERT outperforms other NLP algorithms and baseline model (without integrating text description) because of its dynamic word embedding from built-in self-attention mechanism, which is trained through a tremendous amount of data and contains syntactic and semantic representations. The performance of BERT is relatively better than GPT-3.5 for categorizing violation types, which may be because GPT-3.5 is primarily designed for text generation [43], [49], meaning it predicts the next word based on preceding context. This approach limits its ability to fully understand and classify text without additional tuning. Therefore, fine-tuning is necessary to adapt GPT-3.5 for tasks like violation categorization. In addition, the precision of DeepForest is higher than recall, because the tree-based model pays more attention to the majority group–regular violation. However, the model should identify as many crash-related violations as possible. A recall of 0.78 is acceptable, but it is not good enough.

The performance of TabNet under different NLP algorithms is shown in **Table III**. The average AUC and recall of TabNet are higher than that of the DeepForest. It indicates that TabNet performed better than DeepForest in this study. Furthermore, the same pattern as DeepForest is discovered when comparing the model without description and the model with Jaccard, TF-IDF, and fastText. TabNet with BERT reaches the highest recall and AUC in the comparison. Besides, the recall is much higher than the precision in TabNet because it emphasizes more on minorities (i.e., crash-related violations). TabNet sacrifices precision to reach a higher recall, which is acceptable for traffic safety. In summary, the violation description improves the ability of DeepForest and TabNet to identify crash-related violations, as long as the information in it is extracted appropriately. Otherwise, the information in the violation description cannot enhance the performance of classifiers and can have a negative impact.

B. Performance Evaluations

From the above results, BERT is the most suitable NLP algorithm to extract the information in violation description in this study. In this section, eight classification models are selected for comparison. The same dataset with information of violation description extracted by BERT is used. Except for TabNet, the other algorithms do not inherently address data imbalance. The oversampling method (i.e., Synthetic Minority Oversampling TEchnique) is applied to address the data imbalance issue. SMOTE can help traditional statistical models capture the underlying structure and distribution of the minority class more effectively, potentially leading to better decision boundaries [55], [56]. TabNet is set with a class weight of five to enable the model to consider the minority.

From the results (Table IV), traditional linear models such as SGD classifier and Linear SVM exhibit relatively poor performance, likely due to their limited ability to capture complex, nonlinear relationships in the data. These models assume a linear decision boundary, which may not be sufficient for distinguishing crash-related violations from regular violations, especially when feature interactions play a critical role. While they achieve moderate recall, their low precision and overall AUC indicate a high rate of false positives, making them less reliable for practical enforcement

TABLE IV
PERFORMANCE EVALUATION UNDER DIFFERENT CLASSIFICATION MODELS

	Accuracy	Recall	Precision	F2 score	AUC
SGD	0.7539	0.7904	0.0947	0.3202	0.7716
Linear SVM	0.7885	0.7426	0.1038	0.3329	0.7663
DT	0.9697	0.6512	0.5100	0.617	0.8156
LightGBM	0.9858	0.6560	0.8649	0.6893	0.8263
CatBoost	0.9871	0.6560	0.9103	0.6953	0.8270
XGBoost	0.984	0.6907	0.7714	0.7055	0.8421
DeepForest	**0.9908**	0.7814	**0.9106**	**0.8042**	0.8895
MLP	0.7885	0.7426	0.1038	0.3329	0.7663
TabNet	0.9717	**0.8861**	0.5459	0.7878	**0.9658**

applications. Tree-based models, including DT, LightGBM, CatBoost, and XGBoost, demonstrate significant improvements in predictive performance. The structured nature of decision trees allows them to capture nonlinear dependencies more effectively than linear models. However, the performance varies across tree-based approaches. While boosting models such as LightGBM and CatBoost leverage gradient-based optimization to refine predictions, DeepForest, which employs a cascade forest structure, outperforms them by introducing hierarchical feature extraction. Unlike boosting models that rely on iterative error correction, DeepForest enhances feature representations through multi-layered ensembles, allowing it to capture more complex patterns in the data. This architecture enables DeepForest to achieve higher precision and recall compared to traditional boosting approaches. Among all models, TabNet achieves the highest recall and AUC, making it the most suitable candidate for crash-related violation prediction in this study. By integrating an attention mechanism with instance-wise feature selection and self-supervised learning, TabNet dynamically prioritizes relevant features, reducing the reliance on manual feature engineering. Unlike tree-based models that require structured feature transformations, TabNet directly processes raw tabular data, preserving feature interactions without predefined assumptions. This enables it to outperform traditional models while maintaining interpretability. The combination of feature selection and deep learning mechanisms makes TabNet a robust and efficient approach for identifying crash-prone violations in an enforcement setting.

Although TabNet incorporates several mechanisms to mitigate overfitting, including attentive feature selection and built-in regularization techniques such as instance-wise dropout and sparsity constraints, overfitting remains a concern, particularly given its high recall. To further evaluate its generalizability, future work will explore out-of-sample predictions [57], [58], [59] by testing the model on entirely unseen datasets from different temporal periods, geographic regions, and enforcement contexts. This will help determine whether TabNet's learned feature representations remain stable and reliable across varying conditions. Balancing recall and precision is key to optimizing enforcement resource allocation. Although TabNet's high recall improves detection of crash-prone violations, its higher false positive rate can misclassify low-risk cases. Enforcement agencies can use the model as a preliminary screening tool, followed by secondary verification—using additional risk indicators, officer assessments, or historical records—to ensure decisions rely on human expertise. The model is meant to complement, not replace, law enforcement judgment. Moreover, integrating extra dataset, such as roadway conditions and traffic patterns, can enhance predictive reliability, and future research may also explore ensemble learning to better balance recall and precision.

C. Model Interpretation

While deep learning models achieve excellent predictive accuracy, their "black box" nature limits interpretability. To better understand TabNet's interpretability, we compared its learned feature importance with the significant variables derived from a statistical model with random parameters. This type of statistical model often provides more reliable insights by capturing unobserved heterogeneity [15], [17], [18], [19]. In this study, the response variable, representing regular violations and violations leading to crashes, is binary. Therefore, we adopted a random parameter binary logit model. Initially, we assessed the random effects of each variable individually by specifying them as random parameters and examining whether their standard deviations were statistically significant. Based on these results, we specified the variables with significant random effects as random in the final model estimation, assuming their parameters follow a normal distribution [17], [60], while the remaining variables were treated as fixed. Given the computational intensity involved in estimating such models [15], [17], we managed the data size by employing under-sampling rather than oversampling.

The left side of Figure 7 illustrates instance-wise feature importance (first 30 observations) derived from TabNet's attention mechanism, where each observation assigns different weights to features, influencing its predictive output accordingly. By averaging local feature importance, the global feature importance is shown on the right side. TabNet processes categorical features using embeddings and attentive feature selection, avoiding the one-hot encoding required by the random parameter logit model. In the global feature importance analysis (Figure 7-right), a variable is flagged as significant if any of its categories are significant in the statistical model. Inevitably, some discrepancies remain—for example, in variables such as race, color, vehicle type, visibility, precipitation, and whether the violation involved alcohol inspection. Data-driven approaches prioritize predictive accuracy, whereas its interpretability is concerned with understanding why the model makes specific predictions for individual observations [8], [23]. A more detailed interpretability analysis, grounded in advanced statistical modeling, is essential for achieving a reliable understanding of these factors. Future research could explore advanced statistical models [19], [20], [61] to examine marginal effects, or hybrid statistical–deep learning frameworks [62], [63], which offer a promising balance between predictive performance and interpretability.

It is noted that the most influential feature in TabNet is "violation types" derived from textual descriptions, which is

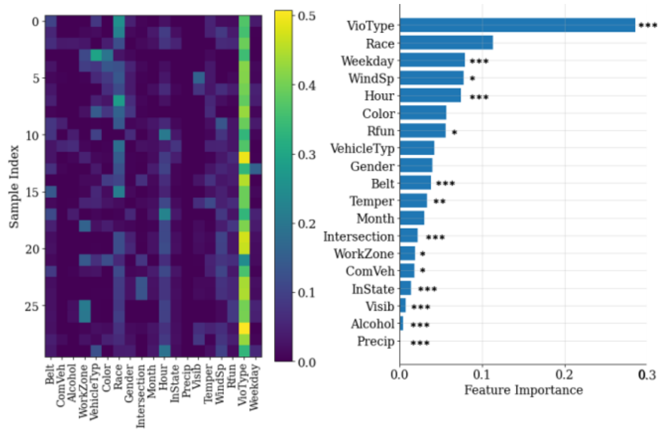

Fig. 7. Sample-wise feature importance (left) and average global feature importance (right) from TabNet. The asterisk (*) denotes significance levels from random parameters binary logit model: * $p < 0.05$ (statistically significant), ** $p < 0.01$ (highly significant), and *** $p < 0.001$ (very highly significant).

also a highly significant variable in the random parameter statistical model. This underscores the value of integrating unstructured textual data into the predictive framework. While the "Alcohol" feature is undeniably crucial for identifying violations that may lead to crashes, its limited representation within the dataset constrains its predictive utility. To elaborate, out of 69,209 records, only 28 involve alcohol suspension, all of which result in regular violations without crashes. Consequently, the data-driven model is unable to leverage this information to differentiate between crash-related and non-crash-related violations. This limitation is expected to be mitigated as more data become available, thereby reducing data bias.

V. Conclusion

The model for predicting individual violations that result in crashes is essential for enhancing traffic safety and helping police officers make informed discretionary decisions. Despite its importance, past studies have not addressed this specific predictive need. Additionally, many existing crash prediction models overlook the value of unstructured data, such as violation descriptions, which may improve prediction accuracy.

This paper considers four NLP algorithms and a LLM framework, including Jaccard, TF-IDF, fastText, BERT, and GPT-3.5 to process the violation description and convert them into structured data. After text processing, eight classification models—including linear models (SGD, SVM with a linear kernel), tree-based models (DT, LightGBM, CatBoost, XGBoost, and DeepForest), and neural network-based models (MLP)—were adopted to evaluate the proposed framework. The results show that incorporating violation descriptions significantly enhances the model's ability to distinguish crash-prone violations, but only with an appropriate NLP algorithm or LLM framework. Otherwise, improperly processed textual data can negatively impact classification accuracy. The performance comparison indicates that BERT and GPT-3.5 improve the performance of DeepForest and TabNet, leveraging self-attention mechanisms to accurately extract valuable information from violation descriptions. Notably, BERT outperforms GPT-3.5 for embedding tasks such as text similarity, likely because LLM may require fine-tuning to achieve better performance. In contrast, Jaccard, TF-IDF, and fastText not only show lower predictive performance but also perform worse than models that do not incorporate textual descriptions. This suggests that improper text processing can introduce noise and degrade classification accuracy rather than providing meaningful improvements.

Next, eight data-driven approaches are used to compare the predictive performance under the same dataset containing information of violation description extracted by BERT. TabNet achieves the best performance in terms of recall and AUC. Its accuracy, recall, and AUC reach 0.9717, 0.8861, and 0.9658, respectively. TabNet's attention mechanism is capable of capturing key features and representations, which reduces the workload from feature engineering. In terms of interpretation, TabNet provides instance-wise feature importance, although its insights do not fully align with those from the random parameter binary logistic model. Nevertheless, it still highlights key factors that drive prediction performance. Our study introduces a comprehensive framework that effectively integrates unstructured data and enhances the accuracy of predicting individual violations that may lead to crashes, offering law enforcement a practical tool for assessing crash risk.

There are several limitations worth mentioning. First, the traffic violation data is solely recorded in the State of Maryland, which may introduce site bias. Expanding the dataset to include records from different locations and sources would improve its representation and quality, also helping to address the issue of sparse representation from the data perspective to certain degree. Second, although we mitigated data imbalance through sampling and class weighting, resampling inevitably alters the underlying distribution. Future work should therefore conduct multiple trials to minimize any unintended bias [64]. Third, although the model has already reflected the temporal information with input variables, it is still possible to further improve the model performance if the data is divided over different time periods, such as peak hour or non-peak hour, and models are built separately for these periods. Finally, while data-driven approaches are powerful, their inherent limitations—such as being non-transparent—can be further addressed by incorporating state-of-the-art statistical models to enhance the robustness and reliability of insights [19], [20], [61] or by adopting a hybrid statistical-deep learning framework [62], [63] to achieve a balanced approach between interpretability and predictive performance.

Appendix

Tables V and VI.

TABLE V
DESCRIPTIVE STATISTICS OF CATEGORICAL FEATURES

Variables	Definition	Attributes	Count	Per. (%)
Crash	if the violation is crash-related	Yes	2,148	3.10
		No	67,061	96.90
Belt	if the seat belts are in use	Yes	67,688	97.80
		No	1,521	2.20
ComVeh	if the violated vehicle is a commercial vehicle	Yes	1,593	2.30
		No	67,616	97.70
Alcohol	if the violation included an alcohol-related suspension	Yes	28	0.04
		No	69,181	99.96
WorkZone	if the violation is in the work zone	Yes	47	0.07
		No	69,162	99.93
Intersection	if the violation is in the intersection	Yes	25,402	36.70
		No	43,807	63.30
Instate	if the driver's license is issued in MD	Yes	65,388	94.48
		No	3,821	5.52
Gender	gender of the driver	M	48,111	69.52
		F	21,098	30.48
VehType	type of vehicle	Passenger Car	64,048	92.54
		Motor	611	0.88
		Truck	3,520	5.09
		Bus	19	0.03
		Other	1,011	1.46
Color	color of vehicle	white	24,887	35.96
		black	15,409	22.26
		blue	7,380	10.66
		red	6,450	9.32
		yellow	3,161	4.57
		green	1,772	2.56
		other	10,150	14.67
Race	race of the driver	Black	21,446	30.99
		White	18,924	27.34
		Asian	3,639	5.26
		Hispanic	19,374	27.99
		Native American	22	0.03
		Other	5,804	8.39
VioType	categories of traffic violation	issue of document	19,363	27.98
		disobeying TCD	16,558	23.92
		unsafe speed	8,839	12.77
		issue of vehicle parts	7,921	11.45
		reckless driving	6,248	9.03
		DWI	3,060	4.42
		distraction driving	2,789	4.03
		failure to yield	2,627	3.80
		illegal parking	1,052	1.52
		failure to wear a seat belt	752	1.09
Rfun	functional classification of the road where issued violation	1- interstate	4,105	5.93
		2- freeway	3,072	4.44
		3- principal arterial	37,910	54.78
		4- minor arterial	11,864	17.14
		5- major collector	9,885	14.28
		6- minor collector	2,373	3.43

TABLE VI
DESCRIPTIVE STATISTICS OF CATEGORICAL FEATURES

Variables	Definition	Mean	Std	Min	Max
Month	month of the traffic violation	5.30	3.77	1.00	12.00
WeekDay	day of the week of the traffic violation	3.91	1.86	1.00	7.00
Hour	hour of the traffic violation	12.73	7.53	0.00	23.00
Temper	hourly temperature at the time of the violation	50.91	16.50	10.00	96.00
Precip	hourly precipitation at the time of the violation	0.00	0.01	0.00	1.28
Visib	visibility at the time of the violation	9.52	1.59	0.21	10.00
WindSp	hourly wind speed at the time of the violation	4.01	3.82	0.00	27.00

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